

Mission Control Interview – Operational Protocols & Communication (Astronaut)

Overview

This document captures firsthand insights into mission control operations, communication protocols, and decision-making practices from the perspective of an astronaut. This information was gathered through an interview that we were able to conduct with **one** astronaut. This information reflects lived experience, operational authority, trust structures, and real-time decision making during flight operations.

Lived Experience

Personal Culture Recount

- The motto over the mission evaluation room at JSC was, *in God we trust, all others bring data*.
- There was never any money to go do the studies to get the data.
- As an astronaut when they would set up the NESC, they got themselves a big budget, and they could hire the best talent in industry, academia, and anywhere else in the world to go and study a problem that is dominated by a chief engineer at the centers that work for the NESC, rather than being submerged under all this program management.
- The chief engineer would go off and study things and come back and give some conclusions about whether the Astronauts needed to stop and fix it or whether it's okay to go.
- In 10 years the NESC had done over 500 in-depth studies about NASA issues, and NASA's had a great safety record ever since then.

Challenger & “Death Star” Risk

- My second flight was supposed to be on the Challenger in May of 86, and we were to fly the first Centaur, which was used on unmanned rockets.
- The chief astronaut called it the Death Star because there was a single bulkhead between the oxygen and liquid oxygen, liquid hydrogen, and it launched.
- They had to pump in extra helium for the structure, and they were having a lot of failures on the valves that put in the helium.

- If you over-pressurized it, it would crack the bulkhead.
- There were other issues, like if you had to bring it home because you couldn't get it out, because the hydrogen went out one side and the oxygen the other, there's just a very slight impingement on the wings, and it caused the shuttle to rotate, and you would use up all your RCS fuel.
- Or another issue where if you did a return alongside aboard, you would still have a few pounds of hydrogen in there because you wouldn't have time to get it out.
- Hydrogen will leak and when you have a payload bay full of wires which have been stepped on all it takes is just a little spark for things to go very bad.
- Our astronaut was the pilot on that, and so I spent my life before going back to the Air Force, building a briefing that says, No. We're not calling this up.

Decision Making

Launch Authority

- Our astronaut once went out to Vandenberg because they were launching a mission, a Titan, and some issues came up and the Air Force was running the countdown, and they were pressing ahead with the launch without getting any clarifications.
- So our astronaut delayed the launch but was entirely vetoed as he was not a very favored guy in the Air Force.

Personal Preparation

- Our astronaut was involved in the first flight to fly a 49.5-degree inclination, while everybody else had flown due east at 28.5.
- In this new flight incline mission they did not have a good downrange abort site.
- Our astronaut went to the flight director, and I said, Flight, I want to get some sim time to qualify a downrange abort site.
 - The Flight Director replied asking "Why in the hell would you do that?"
- Our astronaut responded by saying "if we lose two engines before we get a single engine abort capability, I'm going to need a place to land.
 - The Flight Director said "That's an unrealistic thing. Get the hell out of my office. You're wasting my time."
- So our astronaut went to the simulator anyway and identified a downrange abort site and wrote the hexadecimal code on his knee pad.
- They ended up needing the abort site.

- After the flight, our astronaut went to the flight director, and told him the story, and they both had a big laugh.

Communication

Shifting Authority

- During launch, as soon as you lift off, control shifts from KSC to Houston.
- There's only one person in charge: Houston and they are in charge all the way until you land.
- After landing, then KSC takes over.
- I found the discipline in the mission control room was excellent, it was very clear.

Prioritization & Training

- Mission Control will give the astronauts all the assets, like 6, 8, 10, 40s and the astronauts are expected to be equipped to prioritize what they will need.
- The required knowledge requirements for astronauts comes with a lot of training and the mission control will help you and tell you whether something is the most sensitive.
- There are times where even though you've never done something before, as an astronaut you trust your mission officer to be knowledgeable enough to know that you as an astronaut have to do it.
- Everybody trusts you and all the training builds that trust in yourself.

Engine Failure & “Limits to Inhibit”

- Our astronaut was a part of the only flight to lose an engine going to orbit.
- After they did an abort to orbit, the booster officer saw something in another engine that looked like the same failure that had caused the engine to shut down.
- Their booster officer had them go Challenger Houston, limits to inhibit
- They had never used that switch because it turns off all the safety systems on the remaining two engines.
- But they made it to orbit because they trusted them, and the communication was very clear.

- There were three temperature sensors, and they were failing hot, and then two of them failed hot. They shut the engine down, and that's what caused the first engine to fail.
- The booster officer saw another sensor in one of the other two engines that was cycling hot, and she had us turn it off.
- So the astronauts had no protection on the two remaining engines.